

Course Code: CS1081		
Name of the Programme: BCA (Hons.)		
Name of the Course: Programming with C		
Semester: I		
Course Credit	No. of Hours Per Week (L + T + P)	Total No. of Teaching Hours
4	3:0:2	75
Course Objectives:		
a) Introduce the fundamental principles of programming and develop essential problem-solving skills.		
b) Build proficiency in core C programming constructs, including control structures, functions, arrays, pointers, and strings.		
c) Enable students to write, debug, and optimize efficient C programs through practical implementation.		
d) Lay a strong foundation for exploring advanced computing topics and further programming languages.		

Course outline (Syllabus of the course)

Module - 1 :Introduction to Programming and C Language Basics	No. of Hours
	9 Hrs.
Introduction to Programming, History and features of the C language, Structure of a C program, Compilation and Execution of a C program, Basic syntax: keywords, identifiers, constants, variables, and data types, Operators and expressions, Input and output functions: printf (), scanf().	
Module - 2 : Control Structures Topics	No. of Hours
	9 Hrs.
Decision-making statements: if, if-else, nested if-else, switch, Looping statements: while, do-while, for, break, continue and go to statements.	
Module - 3 : Functions and Arrays	No. of Hours
	9 Hrs.
Functions: declaration, definition, and calling, Function types: with/without arguments, with/without return value, Storage classes, Recursion, Arrays: one-dimensional and two-dimensional arrays.	
Module - 4 : Pointers and Strings	No. of Hours
	9 Hrs.
Introduction to pointers, Pointer arithmetic, Pointers and arrays, Pointers to functions, Strings: declaration, initialization, and string handling functions.	

Module - 5 (Structures, Unions, and File Handling)	No. of Hours 9 Hrs.
Structures: definition, declaration, and initialization, Arrays of structures, Pointers to structures, Unions, File handling: file operations, reading from and writing to files.	

Course Outcomes:
a) Understand the fundamental concepts and basic syntax rules of C programming.
b) Analyze decision-making and looping constructs, and apply string manipulation techniques to solve programming problems.
c) Apply functions, arrays, and pointers, including dynamic memory allocation, to develop modular and efficient code.
d) Create user-defined data types such as structures and unions to organize and manage complex data effectively.

Text Books
1 Balagurusamy, E. (2019). <i>Programming in ANSI C</i> (8th ed.). McGraw Hill Education. ISBN: 978-9351343202
2 Kanetkar, Y. (2022). <i>Let us C</i> (19th ed.). BPB Publications. ISBN: 978-8183331630

Reference Books
1 Kernighan, B. W., & Ritchie, D. M. (2015). <i>The C programming language</i> (2nd ed.). Prentice Hall. ISBN: 978-9332549449
2 Schildt, H. (2017). <i>C: The complete reference</i> (4th ed.). McGraw Hill Education. ISBN: 978-0070411838

Laboratory Components	
1	Write a C program to print your name entered by the user.
2	Write a C program to perform arithmetic operations.
3	Write a C program to swap 2 numbers a. without using a temporary variable b. using a temporary variable
4	Write a C program to find the largest of three numbers.
5	Write a C program to print the first 10 Fibonacci numbers.
6	Write a C program to print the following Half Pyramid of Numbers pattern. 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5
7	Write a C program to calculate the factorial of a number.
8	Write a C program to check whether a number is prime.
9	Write a recursive function to find the GCD of two numbers.
10	Write a function to perform matrix addition and multiplication.
11	Write a program to swap two numbers using pointers.
12	Write a program to reverse a string.
13	Write a C program to Find the Frequency of Characters in a String.
14	Write a program to store and display information of a student using structures.
15	Write a program to read and write a file.
16	Write a program to count the number of lines, words, and characters in a file.

Lesson plan

Name of the Programme	BCA(Hons.)	
Title of the Course	Programming with C	
Course code	CS1081	
Semester	I	
Names of the Course Instructor(s)	Prof. Sharath Shetty BR & Dr. K Thenmozhi	
Course credit	No. of hours per week (3+0+2)	Total no. of Teaching hours
04	05 Hours	75 Hours

Session	Module No.	Topic	Pedagogy / Activities	Pre-reading / Reference
1	1	Introduction to Programming	Interactive lecture; think-and-share	Textbook Ch1; online article
2		History & features of C language	Lecture + discussion	Textbook Ch1
3		Structure of a C program	Code walkthrough + board explainer	Textbook Ch2
4		Lab: First C program – print name	Hands-on coding	Presentations and Tutorials
5		Lab: Arithmetic operations	Pair programming	Presentations and Tutorials
6		Compilation & Execution of C	Live demo using IDE	IDE guide

7	1	Keywords & Identifiers	Lecture; short quiz	Textbook Ch2
8		Constants, Variables, Data Types	Classification activity	Textbook Ch2
9		Lab: Swap numbers (with/without temp)	Debugging practice	Presentations and Tutorials
10		Lab: Largest of three numbers	Instructor demo then solo	Presentations and Tutorials
11		Operators & Expressions	Solve examples in class	Textbook Ch3
12		Input/output ('printf', 'scanf')	Code demo; hands-on	Textbook Ch3
13		Review & Quiz on basics	MCQ and Q&A	Notes till Session 8
14		Lab: Fibonacci (first 10)	Code-along	Presentations and Tutorials
15		Lab: Half-pyramid pattern	Challenge task	Presentations and Tutorials
16	2	Decision making: if	Lecture; whiteboard traces	Textbook Ch4
17		if-else & nested if-else	Think-pair-share; coding demo	Textbook Ch4
18		switch statement	Lecture; case studies	Textbook Ch4

19	2	Lab: Factorial (loops)	Hands-on	Presentations and Tutorials
20		Lab: Prime number check	Pair programming	Presentations and Tutorials
21		Looping: while	Interactive demo	Textbook Ch5
22		Looping: do-while	Lecture; student exercises	Textbook Ch5
23		Looping: for	Live coding	Textbook Ch5
24		Lab: GCD (iterative)	Guided coding	Presentations and Tutorials
25		Lab: Patterns with nested loops	Creative task	Presentations and Tutorials
26		break, continue, goto	Debate: use vs misuse	Textbook Ch5
27		Combined control structures	Problem-solving workshop	Handout
28		Review & mini-test	Quiz; clarify doubts	Module 2 notes
29		Lab: Menu-driven arithmetic ops	Debug & run	Presentations and Tutorials
30		Lab: Code review & debugging	Peer review	Lab codes Module 2

31	3	Functions: declaration & definition	Lecture; code snippets	Textbook Ch6
32		Function calling & parameter passing	Dry-run exercises	Textbook Ch6
33		Function types (args/return)	Table activity	Textbook Ch6
34		Lab: Factorial via recursion	Hands-on	Presentations and Tutorials
35		Lab: Recursive GCD	Pair program	Presentations and Tutorials
36		Storage classes	Lecture; quiz	Textbook Ch6
37		Recursion concepts	Visualization; call stack demo	Textbook Ch6
38		Arrays 1D	Lecture; board diagrams	Textbook Ch7
39		Lab: Matrix addition	Group task	Presentations and Tutorials
40		Lab: Matrix multiplication	Group task	Presentations and Tutorials
41		Arrays 2D	Matrix example	Textbook Ch7
42		Arrays & functions	Pass array demo	Textbook Ch7

43	3	Module 3 review	Q&A	Module 3 notes
44		Lab: Array operations (search/sum)	Individual	Presentations and Tutorials
45		Lab: arrays & functions	scenarios	Presentations and Tutorials on arrays
46	4	Pointers: basics & declaration	Lecture; memory diagram	Textbook Ch8
47		Pointer arithmetic	Hands-on tracing	Textbook Ch8
48		Pointers & arrays	Code demo	Textbook Ch8
49		Lab: Swap numbers using pointers	Hands-on	Presentations and Tutorials
50		Lab: Reverse a string	Pair programming	Presentations and Tutorials
51		Pointers to functions	Advanced demo	Textbook Ch8
52		Strings: declaration & init	Lecture; memory illustrate	Textbook Ch9
53		String handling functions	Live coding	Textbook Ch9
54		Lab: Frequency of characters	Individual task	Presentations and Tutorials
55		Lab: Pointer-based array ops	Group coding	Presentations and Tutorials

56	4	Module 4 review & quiz	Quiz; discussion	Module 4 notes
57		Lab: String manipulation tasks	Challenge lab	Presentations and Tutorials
58	5	Structures: definition & declaration	Lecture; sample code	Textbook Ch10
59		Initialization & accessing members	Code demo	Textbook Ch10
60		Arrays of structures	Diagram activity	Textbook Ch10
61		Nested & pointers to structures	Hands-on	Textbook Ch10
62		Unions: concept & usage	Lecture; compare with structs	Textbook Ch10
63		File handling basics	Lecture; open/close file	Textbook Ch11
64		Lab: Student info using struct	Hands-on	Presentations and Tutorials
65		Lab: Array of structures record system	Implementation with real life examples	Presentations and Tutorials
66		Writing to files	Live coding	Textbook Ch11
67		Reading from files	Live coding	Textbook Ch11
68		Random access in files	Demo with fseek	Textbook Ch11

69	5	Lab: Pointers to structures operations	Pair programming	Presentations and Tutorials
70		Lab: Read & write file	Hands-on	Presentations and Tutorials
71		File error handling	Case studies	Textbook Ch11
72		File error handling	Case studies	Textbook Ch11
73		Error Handling	Debugging exercise	Textbook
74		Lab: Review of all sessions	Git hub	All lab codes
75		Lab: Comprehensive lab assessment	Practical and viva	All lab codes

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component
- CIE 01 a) MCQ	10%
- b) Trace the program for errors	10%
- CIE 02 Pen & Paper (theory)	25%
- CIE 03 a) Lab Exam	15%
- b) Viva	5%
- c) Record	5%

Semester End Exams (SEE)	
Mode of Exam	Theory
Weightage of exam	30%

Course Outcomes- Program Outcomes Matrix

Sl.NO	CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
1	CO1	3	2	2	–	–	1	–	–
2	CO2	3	3	3	–	–	2	–	–
3	CO3	3	3	3	–	–	2	–	–
4	CO4	2	1	2	3	3	3	–	–

Rubrics for evaluation of CIE (separate for each CIE component)

Programme Outcome	1. Application of Knowledge Apply knowledge from the core/interdisciplinary areas and develop skills to solve complex real-world problems. 2. Critical Thinking & Creativity Analyse and evaluate the existing situations, practices and evidence to develop innovative multi-perspective solutions. 3. Digital Transformation Instil an ability for confident, critical and responsible use of digital technologies for learning, work and participation in society. 4. Communication & Interpersonal Skills Develop skills to effectively express thoughts and ideas both orally and in writing to people from all sections of the society.
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	5. Leadership & Collaborative Teamwork Exhibit skills to set goals and guide people to align effectively in building a team to achieve the goals. 6. Lifelong Learning Engage in continuous self-directed learning to adapt to evolving technologies, societal changes and emerging global challenges.					
Total Marks	70					
CIE components	CO-PO	Excellent(10)	Very Good (7-9)	Good (4-6)	Average (1-3)	Needs Improvement (0)
CIE-1 Graded Component 1 (Quiz) Weightage: -10	CO1 PO1-PO3	All answers correct, demonstrating deep understanding of linux and shell Programming fundamentals and concepts with clear explanations	Most answers correct with minor errors, showing good understanding	Some correct answers with partial understanding and notable errors	Few correct answers with significant gaps in understanding	No correct answers or no submission
CIE components	CO-PO	Excellent (10)	Very Good (7-9)	Good (4-6)	Average (1-3)	Needs Improvement (0)
CIE-1 Graded Component 2 (Debugging) Weightage: 10	CO1 PO1, PO2,PO3, PO6	Code is fully correct, optimized, and well-commented. Handles edge cases and found exact logic	Minor mistakes or inefficiencies. Logic mostly correct. Minor edge cases missed.	Basic logic implemented. Some errors or incomplete handling.	Major logic flaws. Code partially runs but is incomplete or error-prone.	No submission or irrelevant work

CIE components	CO-PO	Continuous Internal Evaluation-2 (25 Marks)				
CIE-2 Mid Sem Examination (Theory) Weightage: 25	CO1, CO2, CO3 PO1-PO3, PO6	Mid Sem Examination (Theory)				
		Mid Sem Assessment Pattern: 25 Marks				
		Sl#	Content			
		Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)				
		1	5 Questions of 1 mark each			5
		Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)				
		2	4 Questions of 5 marks each with 2 or 3 sub divisions possibly			20
CIE components	CO-PO	Excellent (10)	Very Good (7-9)	Good (4-6)	Average (1-3)	Needs Improve ment (0)
CIE-3 Lab Weightage: 15	CO1, CO2, CO3, CO4	Refer Table 1				
CIE-3 VIVA Weightage: 5	PO1-PO6	Refer Table 2				
CIE-3 Records Weightage: 5		Excellent (5)	Very Good (4-3)	Good (2)	Average (1)	Needs Improve ment (0)
	CO1, CO2, CO3, CO4 PO1-PO7	Code is fully correct, optimized, and well-commented. Handles edge cases	Minor mistakes or inefficiencies. Logic mostly correct. Minor edge cases missed.	Basic logic implemented. Some errors or incomplete handling.	Major logic flaws. Code partially runs but is incomplete or error-prone.	No submission or irrelevant work

Table 1:

Lab Write up Execution Rubrics (15 Marks)

Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor	Marks	Remarks
1	Understanding of problem statements (5 Marks)	Observations	Student exhibits a thorough understanding of the problem statements and Applies appropriate data structure and Python knowledge to devise a solution. (5 M)	Student has sufficient understanding of the problem statements and applies appropriate data structure and python knowledge to devise a solution. (<4 M and >=1 M)	Student does not have a clear understanding of the problem statements and is unable to apply appropriate data structure and python knowledge to devise a solution. (0 M)		

2	Execution (5 Marks)	Observations	Student demonstrates the execution of the program with appropriate data structure. All test cases are handled with appropriate validations. (5 M)	Student demonstrates the execution of the program with appropriate data structure with only a few test cases handled. (<4 M and ≥ 1 M)	Student has not executed the program or the code fails to demonstrate correct use of data structure. (0 M)		
3	Results and Documentation (5 Marks)	Observations	Documentation with appropriate comments and output is covered in manual. (5 M)	Documentation with only few comments and only few output cases is covered in manual. (<4 M and ≥ 1 M)	Documentation with no comments and no output cases is covered in manual. (0 M)		

Table 2 : Viva Voice Rubrics (5 Marks)

Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor	Marks	Remarks
1	Conceptual Understanding (5 Marks)	Viva Voce	Explains thoroughly the core concepts and principles of data structure used in corresponding lab as well as python constructs. (5 M)	Adequately explains the core concepts and principles of data structure used in corresponding lab as well as python constructs with some understanding . (<5 M and >=1 M)	Unable to explain the core concepts and principles of data structure used in corresponding lab as well as python constructs. (0 M)		

Student list

Section - A

Sl. No.	USN No.	Student name

Section - B

Sl. No.	USN No.	Student name